Result 1 (identity pair endpoints). [Topology foundations, identity inclusion of pairs]
Let X be a topological space. Let P be two nested subsets of X with inclusion evidence. Then

$$\mathrm{id}_P\,.\mathrm{source} = P \text{ and } \mathrm{id}_P\,.\mathrm{target} = P.$$

Result 2 (composed pair endpoints). [Topology foundations, composition of inclusions of pairs]
Let X be a topological space. Let Q and R be elements of PairInclusion. Assume Q.target $= R$.source (`middle`). Then

$$(R \circ Q)\,.\mathrm{source} = Q.\mathrm{source} \text{ and } (R \circ Q)\,.\mathrm{target} = R.\mathrm{target}.$$

Result 3 (pair source identity). [Topology foundations, source identity law for inclusions]
Let X be a topological space. Let Q be a certified inclusion between pairs in X. Then

$$(Q \circ \mathrm{id}_{Q.\mathrm{source}}) = Q.$$

Result 4 (pair target identity). [Topology foundations, target identity law for inclusions]
Let X be a topological space. Let Q be a certified inclusion between pairs in X. Then

$$(\mathrm{id}_{Q.\mathrm{target}} \circ Q) = Q.$$

Result 5 (pair composition associativity). [Topology foundations, associativity of inclusions]
Let X be a topological space. Let Q, R and S be elements of PairInclusion. Assume Q.target $=$ R.source (`first`). Assume R.target $= S$.source (`second`). Then

$$(S \circ (R \circ Q)) = ((S \circ R) \circ Q)\,.$$

Result 6 (identity pair continuous maps). [Topology foundations, underlying maps of identity inclusions]
Let X be a topological space. Let P be two nested subsets of X with inclusion evidence. Then

$$\text{ambient continuous map}(X, \mathrm{id}_P) = \mathrm{id}_{P.\mathrm{ambient}} \text{ and } \text{relative continuous map}(X, \mathrm{id}_P) = \mathrm{id}_{P.\mathrm{relative}}\,.$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.